Group Assignment
Grading Rubric

Data Driven Decision Making
EXAM03 Data Science
CU75073V3

HBO-ICT
2025/2026

UNIVERSITY OF APPLIED SCIENCES

This documents contains the rubric that will be used for EXAM03: the group assignment. In this exam, four learning outcomes are tested using a portfolio of Jupyter Notebooks. This portfolio consists strictly of **three sequential Jupyter Notebooks** (one per iteration). These notebooks must function as standalone documents where code, visualizations and markdown narrative are integrated.

LO	Description
6.4C	You produce a final report and present this to customer
6.4D	You review the data science process and you determine, and also report, lessons learned
7.3Q	As a team you can communicate your research in an organized way, appropriate for the audience
7.3P	Students can present their project, the content of their portfolio and their process considerations in a sound way making plausible the equal contribution of each project member to the project

Grading protocol

Grading process

- Student work for EXAM03 is graded based on the rubric for EXAM03 of that study year.
- Student work is graded based on the rubric covering the learning outcomes for EXAM03.
- Only work that has been handed in on Learn before the deadline will be graded.
- Student work is checked for completeness; student work is checked for the prerequisites. If the work does not meet the prerequisites, the work is automatically graded with a 1,0.
- Student work is graded on Learn, based on the information contained in this document.

The preliminary cut-off point

A sufficient result (minimum 5.5) must be obtained for EXAM03 to successfully complete the course. The four Learning Outcomes each determine 25% of the final grade. Altogether, the final result should at least be 5.5 to pass EXAM03. Note that LO 7.3P will have an individual grade, based on contribution, effort and personal lessons learned.

The method for embedding feedback into education;

During the course, the classes and workshops will have lecturers present to help you work on your group work and implement the learned knowledge and skills to your assignment. In the final week, the final workshops are reserved for providing a final check on your group work.

The assignment is divided into smaller parts that build on each other. If a student group follows this schedule, they will be on track to successfully complete the exam with a satisfactory result.

Rules for countering assessor effects

- Students submit in groups.
- Students' work will be graded by two assessors.

Each learning outcome is defined by indicators that are graded based on the group documents that have been handed in. The group documents that have been handed in should be relevant for the assignment for that study year.

Five score categories are made for each indicator (connected to the learning outcomes.) For each category, a default is given. That is the score you will receive when you score "up to standards" within this category. If you do "more than required" and exceed expectations compared to the default, but not quite enough to be placed in the next category, you will receive the upper bound score from the category. For "less than is required" compared to the default but still enough for that category, you will receive the lower bound score from that category.

It is important to realize that the descriptions of the categories are meant as mere examples. While it is smart to apply everything that is mentioned in the description, this will not automatically mean that your work belongs to the corresponding category and this will be awarded to you.

The examiners will fill in a score and feedback (if applicable) per LO. The final grade is the average across all these scores; each LO is 25% of the final grade.

N/A (1)	**Insufficient** **(3.3 – 5.4)** **Default: 4.4**	**Sufficient** **(5.5 – 7.0)** **Default: 6.2**	**Good** **(7.1-8.5)** **Default: 7.8**	**Excellent** **(8.6 – 10)** **Default: 9.3**
Missing/ no evidence	Description of when something belongs to this category.	Description of when something belongs to this category.	Description of when something belongs to this category.	Description of when something belongs to this category.
	Feedback (when deviating from default):			

For example:

- If you are doing *barely* or *just enough* for the category *Sufficient*, you will receive a 5.5 for that indicator;
- If you are doing exactly what is expected for the category *Sufficient*, you will receive a 6.2 for that indicator;
- If you are doing *more* than is expected for *Good*, but it is *not quite* enough for *Excellent*, you will receive a 7.8 for that indicator.

Prerequisites

If the work does not meet the prerequisites, the work is automatically graded with a 1,0. Additionally the learning objectives will not be graded.

	Prerequisite
☐	The submission contains the three Jupyter notebook templates provided (one for each iteration) follows the structure, content and guidelines as described in the three provided Jupyter notebook templates.
☐	The content of the notebooks is understandable in terms of spelling, grammar, sentence structure and text ordering
☐	The content of the notebooks is appropriate in terms of the correct application of data science theory in practice (data science *mistakes* will affect your grade)
☐	The notebooks contain the content as prescribed by the exam content descriptions ("Deliverables for Exam03") on Learn for the three iterations
☐	The content is coherent, descriptions match the code and visualizations
☐	The notebooks follow up seamlessly and can be run after another without issues
☐	The submitted content is authentic and unique

Grading

See specific LO grades below.

Student Name	
Final Grade	#,#

[25%] Learning Outcome 6.4C: You produce a final report and present this to customer

Indicators:

1. Clarity, organization and content of the final notebooks
   - The notebooks are well-structured and have a logical flow in terms of presenting the data science process that has been followed using CRISP-DM. Code execution is linear and reproducible.
   - You show understanding of the execution of the technical assignment based on the phases of CRISP-DM.
   - The content of the notebooks is correct in terms of Data Science steps taken
2. Integration of iterations into the final notebooks
   - The notebooks synthesize data science insights and results from all three iterations into a cohesive narrative using appropriate narrative cells
   - You demonstrate how findings or lessons learned from earlier iterations informed later work by using markdown summaries.
3. Presentation of results
   - Visualizations are technically precise, relevant to the business objective, and effectively support and enhance the data science findings and insights.
   - For every iteration, results are communicated in a way that addresses the business objectives of that iteration.

N/A (1)	**Insufficient (3.3 – 5.4) Default: 4.4**	**Sufficient (5.5 – 7.0) Default: 6.2**	**Good (7.1-8.5) Default: 7.8**	**Excellent (8.6 – 10) Default: 9.3**
Missing/ no evidence	Notebooks do not cover the CRISP-DM process. Data science steps taken are incorrect. Data science insights from iterations are mentioned, but connections between them are weak or unclear. Data science insights from visualizations are missing, irrelevant or poorly explained; results do not address objectives.	Notebooks contain description of the CRISP-DM process followed for the iterations, but lacks smooth transitions between CRISP-DM phases Data science steps taken are correct. Data science insights from iterations are mentioned, but connections between them are weak or unclear. Data science insights from the visualizations are adequate, but lack clear alignment with the iterations; results partially address objectives.	Notebooks contain well-structured descriptions of the CRISP-DM process followed, with clear transitions between CRISP-DM phases. Data science steps taken are correct. Data science insights from iterations are synthesized into a cohesive narrative that demonstrates progression. Visualizations are clear and relevant in terms of providing data science insights, results support business objectives.	Notebooks contain exceptionally well-organized description of the CRISP-DM followed, with seamless flow and clear alignment to CRISP-DM phases. Data science steps taken go beyond class material and have been applied correctly. Data science findings and insights are seamlessly integrated; Visualizations are highly effective in terms of providing data science insights; results are directly aligned with the business objectives.
Grade: #,#	**Feedback (when deviating from default):**			

[25%] Learning Outcome 6.4D: You review the data science process and you determine, and also report, lessons learned

Indicators:

1. **Reflection on the data science process**
   - You critically evaluate your approach across all CRISP-DM phases (business understanding, data understanding, preparation, modeling, evaluation).
   - You identify strengths and areas for improvement in your process.
2. **Identification of lessons learned**
   - You clearly articulate lessons learned from each iteration for every phase of CRISP-DM.
   - You provide actionable recommendations for improving future projects.
3. **Discussion of challenges faced**
   - You acknowledge challenges encountered during the project (e.g., data cleaning issues, model limitations) and describe how they were addressed or could be mitigated in the future.

N/A (1)	**Insufficient (3.3 – 5.4) Default: 4.4**	**Sufficient (5.5 – 7.0) Default: 6.2**	**Good (7.1-8.5) Default: 7.8**	**Excellent (8.6 – 10) Default: 9.3**
Missing/ no evidence	Little to no objective reflection on phases of the CRISP-DM process for each iteration; Lessons learned from previous iterations are not focused on the data science process, vague or missing; challenges are not discussed.	Basic objective reflection on the phases of the CRISP-DM process for each iteration; Lessons learned from previous iterations are identified but not well-developed; challenges are briefly mentioned.	Thoughtful and objective reflection on the phases of the CRISP-DM process per iteration; Lessons learned from previous iterations are specific and actionable; challenges are discussed with solutions proposed.	Comprehensive and objective reflection on all CRISP-DM phases per iteration; Lessons learned from previous iterations demonstrate deep insight; challenges are thoroughly analyzed with innovative solutions.

Grade: #,#	**Feedback (when deviating from default):**

[25%] Learning Outcome 7.3Q: As a team you can communicate your research in an organized way, appropriate for the audience

Indicators:

1. **Organization of research**
   - You have described the steps taken in a logical and coherent manner.
   - You have organized the text, code and outputs in a cohesive way.
   - The notebooks can be understood without looking at the raw code (i.e. outputs and markdown should tell the story)
2. **Use of visualizations**
   - You use clear and easily interpretable visualizations to support the content of the portfolio
   - The visualizations are technically precise and are followed/accompanied by a markdown interpretation of the insights
   - You have designed the visualizations carefully, paying attention to clarity, legibility and relevance of the visualizations
3. **Clarity of communication**
   - The language used is clear and appropriate, facilitating proper communication.

N/A (1)	**Insufficient (3.3 – 5.4) Default: 4.4**	**Sufficient (5.5 – 7.0) Default: 6.2**	**Good (7.1-8.5) Default: 7.8**	**Excellent (8.6 – 10) Default: 9.3**
Missing/ no evidence	Steps taken are disorganized or unclear; the research process cannot be followed. Visuals are missing, cluttered, irrelevant, or difficult to interpret. Language is inappropriate for the audience (e.g., overly technical or vague); communication is confusing.	Steps are described in a simple sequence but lack smooth transitions or logical flow. Visuals are included but lack polish or contextual explanation; some visualizations may be unclear or generic. Language is somewhat clear but may be too technical or lack precision for the audience.	Steps are described logically and coherently, with a clear progression that the audience can follow. Visuals are stylistically clear, and enhance understanding of the research process and findings. Language is clear and mostly tailored to the audience; explanations are easy to follow.	Steps are well-organized, with a clear narrative structure tailored to the audience. Visuals are polished, innovative, and significantly enhance understanding of the research work. Language is clear, concise, and tailored to the audience.

Grade:	**Feedback (when deviating from default):**
#,#	

[25%] Learning Outcome 7.3P: Students can present their project, the content of their portfolio and their process considerations in a sound way making plausible the equal contribution of each project member to the project

Indicators:

1. **Evidence of equal contribution**
   - You include evidence of what the different group members have worked on in the final portfolio.
   - Personal lessons learned further clarify your individual learning paths.
   - In the portfolio and in the lessons learned, you indicate each individual's contribution. Each student gets an individual grade for this LO based on this contribution.
2. **Integration of individual contributions into final portfolio**
   - You ensure that the overall writing style, coding style, conventions and formatting of the portfolio is consistent and coherent.
   - The different parts of the portfolio form one cohesive portfolio (within iterations and between iterations).

N/A (1)	**Insufficient (3.3 – 5.4) Default: 4.4**	**Sufficient (5.5 – 7.0) Default: 6.2**	**Good (7.1-8.5) Default: 7.8**	**Excellent (8.6 – 10) Default: 9.3**
Missing/ no evidence	No/little personal contribution evident, imbalance in effort is clear, no variety in tasks; Roles and responsibilities are unclear or unbalanced. Individual contributions are poorly integrated into the final portfolio; Sections feel disjointed or inconsistent in quality/style. Personal lessons learned are not focused on the data science process, or lack reflective level.	Some contribution is clear, slight imbalance in effort. Slight variety in tasks per team member during the overall process. Individual contributions are somewhat integrated but may lack consistency in quality/style across sections of the portfolio. Personal lessons learned are focused on the data science process, show basic reflective level.	Clear evidence of equal contribution among team members with balanced roles and responsibilities; Team members diversified task allocation during the process. Minor areas could improve clarity further. Individual contributions are well-integrated into a cohesive final portfolio with consistent quality/style throughout; Minor areas could improve further cohesion. Personal lessons learned demonstrate clear understanding of specific CRISP-DM phases with thoughtful reflection on individual growth and learning.	Strong evidence of equal contribution among team members with clearly defined roles and seamless collaboration. Team members clearly worked on a variety of tasks during the overall process, diversification of tasks has been made explicitly clear. Individual contributions are seamlessly integrated into a cohesive final portfolio that demonstrates high-quality work throughout. Personal lessons learned show deep, critical reflection on the data science process with concrete examples, demonstrating sophisticated understanding of personal development and future applications.
Grade: **#,#**	**Feedback (when deviating from default):**			